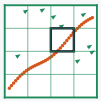
(a) Local encoders $a_\ell = E_\ell(Y_\ell^{\text{in}})$, by data type

irregular: tracks, buoys, SAR drift, L2



e_n tokens

cross-attention per cell
learned query q

$e_n = \text{MLP}[y_n, \sigma_n, \text{type}_n, x_n - x_k, t_n - t]$

$a_\ell(k) = \Sigma_{n \in k} \text{softmax}_{n \in k}(q^\top W_K e_n)\, W_V e_n$

dense gridded: ASIP SIC, gridded PMW



masked strided CNN
(patch-embed + coverage)

only non-empty tiles; mask channel = coverage

forcing: ERA5 $T_{2m}, q_{2m}, u_{10}, v_{10}$



CNN + regrid $\to f$ on $\Omega_{\mathcal{C}}$

(b) Bottom-up pyramid: h_ℓ and ψ_ℓ

$f = E_F(F)$ enters at $\mathcal{C}$

$h_{\mathcal{C}} = U_{\mathcal{C}}(a_{\mathcal{C}}, \Pi^\uparrow h_{\mathcal{M}}, f)$

$\Psi_{\mathcal{C}} \to \psi_{\mathcal{C}}$

$\Pi^\uparrow$

$h_{\mathcal{M}} = U_{\mathcal{M}}(a_{\mathcal{M}}, \Pi^\uparrow h_{\mathcal{H}})$

$\Psi_{\mathcal{M}} \to \psi_{\mathcal{M}}$

$\Pi^\uparrow$

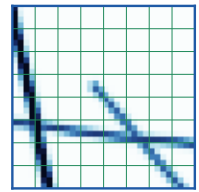
$h_{\mathcal{H}} = a_{\mathcal{H}}$

fused per $\mathcal{C}$ tile for 500-m ASIP; on the fly

(c) $\Pi^\uparrow$ in one $\mathcal{C}$ cell

64 children i ($\mathcal{M}$)

weights α_{ji}



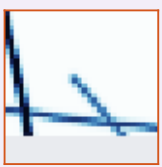
$(\Pi^\uparrow h)(k) = W_O[\Sigma_{i \in \text{ch}(k)} \alpha_{ji} W_V \tilde{h}_i]_{j=1..K}$

$\alpha_{ji} = \text{softmax}_i(q_j^\top W_K \tilde{h}_i / \sqrt{d}), \quad \tilde{h}_i = h(i) + p_i$

q_j: K learned queries ("questions")
p_i: position of the child in the parent
masked children excluded; all empty $\to \varnothing$

(d) Sub-grid parametrization $\Phi_\ell = (\mathcal{S}_\ell, \psi_\ell)$: explicit part vs latent part

$\mathcal{S}_\ell$: physical, explicit, supervised



$\hat{X}_{\mathcal{H}} = \mathcal{G}(Y_{\mathcal{H}}^{\text{tgt}})$, cell of (c)

$\mathcal{S}_{\mathcal{C}}$

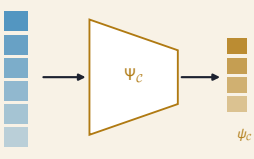


target of $S_{\mathcal{C}}$
in $Z_{\mathcal{C}}$ (generated
by $G_{\mathcal{C}}$): $\mathcal{L}_S$

$\hat{S}_{\mathcal{C}}$: computed on the fly,
no learnable parameter

$m^S = \mathbb{1}[\text{cov} \geq \gamma], \quad \sigma_S^2 = \sigma_0^2 + \sigma_{\text{samp}}^2 / n_{\text{eff}}$

ψ_ℓ: latent, small, estimated for conditioning

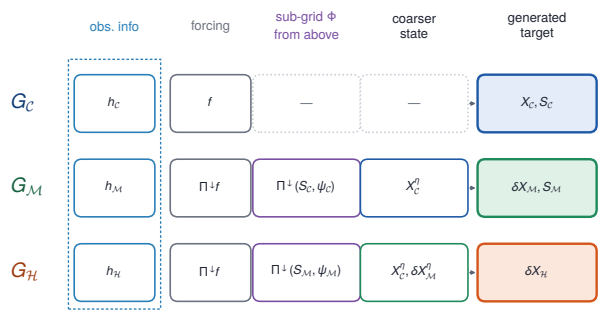


$h_{\mathcal{C}}(k) \in \mathbb{R}^d$

no target in $\mathcal{N}_0$: shaped by the losses of
$G_{\mathcal{M}}, G_{\mathcal{H}}$ that it conditions
$\mathcal{L}_\psi = E\,\|\psi\|^2 + \lambda_\perp \|\text{Cov}(\psi, S)\|_F^2$
in $\mathcal{N}_1$: target $\bar{\Psi}(h^{t+\Delta})$ (EMA)

$\psi_{\mathcal{C}} \in \mathbb{R}^{d_\psi}$, $d_\psi \approx 4$–8

(e) Assembling the contexts c_ℓ



$h_\ell = \varnothing$ without observations ($\mu = 0$): forcing + Φ drive a free simulation; observations add a nudging term

$v_w = v(\cdot \mid \varnothing) + w[v(\cdot \mid h) - v(\cdot \mid \varnothing)]$

coarse obs. do not descend: they act through $X_{\mathcal{C}}$

$X^\eta = X + \eta$: generated coarse state, slightly noised

training: each group replaced by $\varnothing$ with prob. $p_\varnothing$

everything is computed on the fly, per batch